Polynomials

1

a Yes, it is a polynomial.

b Yes, it is a polynomial.

c No, it is not a polynomial.

d Yes, it is a polynomial.

e No, it is not a polynomial.

f No, it is not a polynomial.

g No, it is not a polynomial.

h Yes, it is a polynomial.

2

a

i 7 ii 2 iii 2

b

i 5 ii $-\sqrt{5}$

iii $7\sqrt{6}+5$

c

i 7 ii $\frac{1}{5}$ iii 5

d

i 3 ii $-\frac{7}{10}$ iii 6

Add and subtract polynomials

3

a $2x^3-6x^2-x+4$

b 3

c 2

d Yes

4 $P(x)+Q(x)=5x^2-8x+2$

5 $P(x)-Q(x)=3x^2+x+1$

6 $B(x)-A(x)=6x^2+4x-1$

7 $A(x)+B(x)+C(x)=4x^2-4x$

8 $A(x)-B(x)-C(x)=7x^2-4x-1$

9

a $4x^3-4x^2-10x-4$

b $-5x^3-4x^2-10x-1$

10

a $-12x^2 - 8x - 3$

b $-4x^3 - 9x^2 - 17x - 7$

c $\frac{5}{6}x^2 + \frac{3}{20}x + \frac{1}{4}$

d $-2x^3 + 6x^2 + 7$

e $-7x^2 - 7x + 26$

f $-9x^3 + 4x^2 + 3x - 1$

g $2x^2 - x - 13$

h $-11x^3 + 9x - 2$

i $-2x^3 - 3x^2 + 9x + 5$

j $3x^3 - 8x - 7$

11 $a = 1,\ b = 12$

Multiply polynomials

12

a $x^2 + 7x + 10$

b $x^2 - 5x + 6$

c $ab - 6a + 9b - 54$

d $m^2 - 3m - 18$

e $5m^2 - 7m - 24$

f $42y^2 + 6y - 36$

g $35w^2 + 39w + 10$

h $\frac{1}{9}y^2 + \frac{1}{36}y - \frac{1}{24}$

i $-90y^2 + 26y + 48$

j $9y^2 + 90y + 144$

k $-x^2 - 7x - 10$

l $-5y^2 - 45y - 100$

m $-4y^2 + 12y + 112$

n $-75x^2 + 110x - 40$

13

a $-30x^2 - 42x + 34$

b $10n^2 + 29n + 6$

c $5a^3 + 8a^2 - 2a + 4$

d $u^3 + u^2 - 15u - 20$

e $-25a^3 - 50a^2 - 45a - 20$

f $20c^4 - 15c^3 + 2c^2 + 6c - 4$

g $-12m^3 - 20m^2 + 8$

h $-56x^2 + 24xy - 12x + 36y + 108$

14 True

15 The linear x terms will come from multiplying constants with terms in x. In particular, the x term will be:

$$(-3x)(-4) + (5x)(-5) = -13x$$

So, the coefficient of x will be -13.

16

a 12

b -18

c Yes

17 $(x+6)(x+2)$

Applications

18

a $8x^2+18x+14$ units

b $4x^3+6x^2+3x+8$ units

c $16x^2+20x+18$ units

d $(6x^2+14x+24)$ units

e $(3x^2+24x)$ units

f $(3x^3+8x^2+4x+9)$ units

19 $12x^3+34x^2-18x+6$

20 $42x^2+59xy+20y^2$

21 $8y^3+22y^2+17y+3\ \text{ft}^3$

22 $12x^2+144x+384\ \text{m}^3$

23 $60x-32x^2+4x^3\ \text{in}^3$

24

a $h+9$ in

b $w+8$ in

c $4h+\dfrac{9w}{2}+36$ in

25 $\left(128y^2+48y\right)\ \text{ft}^3$